A monolithic cell-centred finite volume approach for fluid-solid interaction

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Abstract

WIP

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1 Introduction

Over the past four decades, immense progress has been made on methods for solving fluid-solid interaction problems [**? ? ?**]. Modelling approaches for fluid-solid interaction can be classified by:

- **Modelling viewpoint**: Eulerian vs Lagrangian vs mixed Eulerian-Lagrangian;
- **Degree of coupling**: weakly vs strongly coupled;
- **Solver structure**: partitioned vs monolithic.

Eulerian approaches treat both solid and fluid domains as "fluid-like" (e.g. volume-of-fluid [**?**], level-set [**? ?**]), easily allowing for arbitrarily large solid deformations. However, conservative, accurate interface descriptions and the inclusion of advanced solid constitutive laws are challenges. This would make the inclusion of heart valves particularly troublesome, for example. In contrast, Lagrangian (particle-type) approaches treat both domains as "solid-like", e.g. smooth particle hydrodynamics (SPH), material point method (MPM) and the particle finite element method (PFEM) [**? ? ?**]. Although promising for large deformation problems, Lagrangian methods typically come with a prohibitively expensive computational cost; this restriction would be fatal for cardiac xenotransplant simulations, which are already expected to be slow. Finally, mixed Eulerian-Lagrangian approaches treat the solid and interface as Lagrangian and the fluid as arbitrary Lagrangian-Eulerian (ALE). This approach is effective for small interface deformations but will eventually fail for large deformations [**?**]. Consequently, significant attention has been focused on methods that allow the solid to "float over" the Eulerian fluid grid, e.g. immersed boundaries/domains [**? ? ?**], Chimera/overset methods [**? ? ? ?**] and cut-cell methods [**? ? ?**]. Of these approaches, overset methods provide the same large deformation freedom but with the added advantage of maintaining orthogonal interface cells to capture fluid boundary layers.

In cases where strict enforcement of kinematic and dynamic interface continuity conditions is not required, reasonable results can be attained when both the fluid and solid domains are solved once per time step [**?**]. However, this approach is unstable for strongly coupled systems, as is expected in cardiac xenotransplant simulations. In contrast, strongly coupled approaches strictly enforce the interface conditions at each time step. This strong coupling can be achieved in different ways, depending on the solver structure: partitioned vs monolithic.

In partitioned (staggered/two-system) approaches, separate solvers (and potentially discretisations) are used for the fluid and solid [**?**]. Consequently, outer iterations must be performed over the fluid and solid domains at each time step, e.g. using Aitken's acceleration [**? ? ? ?**], quasi-Newton procedures [**? ?**], or Robin-type coupling [**?**]. The main limitations of these approaches can be robustness and efficiency, whereas for highly nonlinear, tightly coupled systems, convergence may not be possible or may be prohibitively slow. This is of concern for a whole heart simulation with multiple deforming structures (valves, chambers, vessels) undergoing complex deformations. In contrast, monolithic approaches treat the entire solid-fluid domain as one system, where the discretised fluid and solid problems are solved concurrently in the same linearised system [**?**]. The primary advantage of monolithic methods is their ability to deal with tightly coupled nonlinear systems, as well as potentially being faster than partitioned approaches (fewer outer iterations). The disadvantages are related to the loss of modularity and the resulting poorly conditioned linear system being more challenging to solve.

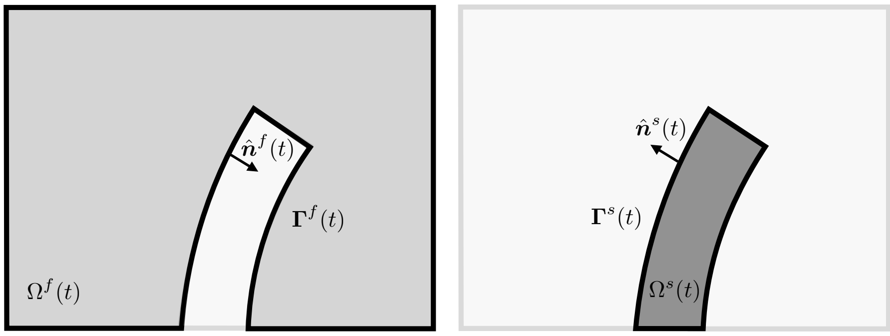
More references:

- **?**] masters thesis, [**?**]
- Wall papers: Gee
- Heil UoM 2004 and 2008
- Tandis thesis
- Greenshields and related uniform FSI papers

2 Mathematical Models

The present work focuses on the time evolution of a system (Figure 1) that occupies moving domain $\Omega(t)$, which is composed of a moving solid domain $\Omega^s(t)$ and a moving fluid domain $\Omega^f(t)$. The boundary of system domain $\mathbf{\Gamma}$ is composed of solid $\mathbf{\Gamma}^s$ and fluid $\mathbf{\Gamma}^f$ regions, and the region where the solid and fluid boundaries coincide is termed the *fluid-solid interface*. The problem consists of determining the time evolution of the system configuration from the initial configuration (Ω_0^s, Ω_0^f, $\mathbf{\Gamma}_0^s$, $\mathbf{\Gamma}_0^f$) to the deformed configuration ($\Omega^s(t)$, $\Omega^f(t)$, $\mathbf{\Gamma}^s(t)$, $\mathbf{\Gamma}^f(t)$) at time t.

(a) Initial configurations of the fluid (left) and solid (right) domains, showing the initial volumes Ω_0, boundaries $\mathbf{\Gamma}_0$ and outward-facing unit normals $\hat{\boldsymbol{n}}_0$



(b) Deformed configurations of the fluid (left) and solid (right) domains at time t, showing the deformed volumes $\Omega(t)$, boundaries $\mathbf{\Gamma}(t)$ and outward-facing unit normals $\hat{\boldsymbol{n}}(t)$

Fig. 1 Representative fluid-solid interaction problem, where superscript f represents fluid domain quantities and superscript s represents solid domain quantities. Image adapted from **?**]

OneraWebsite: https://w3.onera.fr/erc-aeroflex/en/project/strategies-for-coupling-the-fluid-and-solid-dynamics

2.1 Solid Dynamics Governing Equations

Dynamics of the solid region are governed by linear momentum conservation, which can be expressed in Lagrangian form on the deformed configuration as

$$\int_{\Omega^s} \frac{\partial \rho^s \boldsymbol{v}}{\partial t} d\Omega^s = \oint_{\Gamma^s} \hat{\boldsymbol{n}}^s \cdot \boldsymbol{\sigma} \, d\Gamma^s + \int_{\Omega^s} \boldsymbol{f}_b^s \, d\Omega^s \tag{1}$$

where ρ^s is the density, $\boldsymbol{v}^s$ is the velocity vector, $\hat{\boldsymbol{n}}^s$ is the unit outwards-facing normal to surface Γ^s, $\boldsymbol{\sigma}^s$ is the true (Cauchy) stress tensor, and $\boldsymbol{f}_b^s$ is a body force per unit volume, e.g. $\boldsymbol{f}_b^s = \rho^s \boldsymbol{g}$ for a gravity body force, where $\boldsymbol{g}$ is gravitionational acceleration. For notational simplicity, the time dependence notation (t) has been omitted from the moving volumes and surfaces; that is, $\Omega(t) \equiv \Omega$ and $\Gamma(t) \equiv \Gamma$.

The integrals above (Equation 1) are expressed over the as-yet-unknown deformed configuration. These integrals can be reexpressed over the known initial configuration by introducing the deformation gradient $\boldsymbol{F} = \mathbf{I} + (\boldsymbol{\nabla d})^{\mathrm{T}}$ and its determinant as $J = \det[\boldsymbol{F}]$, in terms of the displacement vector $\boldsymbol{d}$ and del operator $\boldsymbol{\nabla}$. Note that the current work adopts the convention that $(\nabla d)_{ij} = {\partial d_j}/{\partial x_i}$, so, to avoid confusion, $\boldsymbol{F}$ and J are expressed in index notation as $F_{ij} = \delta_{ij} + \frac{\partial d_i}{\partial X_j}$ and $J = \frac{1}{6}\varepsilon_{ijk}\varepsilon_{pqr}F_{ip}F_{jq}F_{kr}$, where δ_{ij} is the Kronecker delta and ε_{ijk} is the permutation symbol.

Employing $\boldsymbol{F}$ and J, volume integrals can be expressed over the initial configuration as

$$\int_{\Omega^s} \mathrm{d}\Omega^s = \int_{\Omega_0^s} J \, \mathrm{d}\Omega_0^s \tag{2}$$

From this, J can be interpreted as the ratio of the deformed to the initial volume $J = {\Omega^s}/{\Omega_0^s}$, allowing the current density to be expressed in terms of the initial density as $\rho^s = J\rho_0^s$. Similarly, surface integrals over the deformed configuration can be reexpressed over the initial configuration as

$$\oint_{\Gamma^s} \hat{\boldsymbol{n}}^s \cdot \boldsymbol{\phi} \, \mathrm{d}\Gamma^s = \oint_{\Gamma_0^s} \left(J\boldsymbol{F}^{-T} \cdot \hat{\boldsymbol{n}}_0^s\right) \cdot \boldsymbol{\phi} \, \mathrm{d}\Gamma_0^s \tag{3}$$

where $\boldsymbol{\phi}$ represents an arbitrary scalar, vector or tensor field, and $\cdot$ operator is replaced by a suitable scalar/vector/tensor multiplication operator. Here, $J\boldsymbol{F}^{-T}$ can be seen to rotate and scale the normal vector from the initial to the deformed configuration; this stems from Nanson's relation, $\boldsymbol{\Gamma} = J\boldsymbol{F}^{-T} \cdot \boldsymbol{\Gamma}_0$, which relates the deformed area vector $\boldsymbol{\Gamma}$ to the initial area vector $\boldsymbol{\Gamma}_0$.

With these mappings, linear momentum conservation can be expressed in the so-called *total Lagrangian* form, where integrals are given over the initial configuration:

$$\int_{\Omega_0^s} \rho_0 \frac{\partial^2 \boldsymbol{d}}{\partial t^2} \, d\Omega_0^s = \oint_{\Gamma_0^s} \left(J\boldsymbol{F}^{-\mathrm{T}} \cdot \boldsymbol{n}_0^s\right) \cdot \boldsymbol{\sigma}^s \, d\Gamma_0^s + \int_{\Omega_0^s} \boldsymbol{f}_{b_0}^s \, d\Omega_0^s \tag{4}$$

The body force $\boldsymbol{f}_{b_0}^s$ is per unit *initial* volume, e.g. $\boldsymbol{f}_{b_0}^s = \rho_0^s \boldsymbol{g}$.

The definition of the true stress ($\boldsymbol{\sigma}^s$ in Equations 1 and 4) is given by a chosen mechanical law. Four mechanical laws are considered in this work, as defined in Appendix A of **?**]: linear elasticity (Hooke's law), and three forms of hyperelasticity (St. Venant-Kirchhoff, neo-Hookean, and Guccione).

Three types of boundary conditions for the solid domain are considered here: prescribed displacement, prescribed traction, and symmetry.

2.2 Fluid Dynamics Governing Equations

The dynamics of the moving fluid region are governed by mass conservation and linear momentum conservation. Mass continuity is equivalent to volume continuity for an incompressible material and can be expressed for a moving domain as

$$\frac{d}{dt}\int_{\Omega^f} d\Omega^f \; + \oint_{\Gamma^f} \hat{\boldsymbol{n}}^f \cdot (\boldsymbol{v}^f - \boldsymbol{v}^\Gamma)\, d\Gamma^f = 0 \tag{5}$$

where $\boldsymbol{v}^f$ is the fluid velocity, and $\boldsymbol{v}^\Gamma$ is the velocity of the moving surface Γ^f. The continuity equation simplifies to

$$\oint_{\Gamma^f} \hat{\boldsymbol{n}}^f \cdot \boldsymbol{v}^f \, d\Gamma^f = 0 \tag{6}$$

by using the geometric (space) conservation law

$$\frac{d}{dt}\int_{\Omega^f} d\Omega^f \; - \oint_{\Gamma^f} \hat{\boldsymbol{n}}^f \cdot \boldsymbol{v}^\Gamma \, d\Gamma^\Gamma = 0 \tag{7}$$

Similarly, linear momentum can be expressed in Eulerian form for a laminar, Newtonian, incompressible, isothermal fluid in a moving domain as

$$\frac{d}{dt}\int_{\Omega^f} \boldsymbol{v}^f \, d\Omega^f + \oint_{\Gamma^f} \hat{\boldsymbol{n}}^f \cdot (\boldsymbol{v}^f - \boldsymbol{v}^\Gamma)\boldsymbol{v}^f \, d\Gamma^f = \oint_{\Gamma^f} \hat{\boldsymbol{n}}^f \cdot \boldsymbol{\sigma}^f \, d\Gamma^f + \int_{\Omega^f} \boldsymbol{f}_b^f \, d\Omega^f \tag{8}$$

where $\boldsymbol{f}_b^f$ are body forces per mass, e.g., $\boldsymbol{f}_b^f = g$. The true (Cauchy) stress $\boldsymbol{\sigma}^f$ is defined for a Newtonian fluid as

$$\boldsymbol{\sigma}^f \; = \nu \boldsymbol{\nabla} \boldsymbol{v}^f + \nu \left(\boldsymbol{\nabla} \boldsymbol{v}^f\right)^T - p\mathbf{I} \tag{9}$$

where ν is the kinematic viscosity, p is the kinematic pressure (pressure per density), and $\mathbf{I}$ is the second-order identity tensor.

As was the case for the solid dynamics equations, the integrals above (Equations 6 and 13) are expressed over the unknown deformed fluid configuration and must be expressed over a known configuration. To achieve this, we employ the deformation gradient $\boldsymbol{F}^\Gamma$ for the moving fluid domain; that is, the deformation of the fluid domain is driven by the motion of the fluid-solid interface, where, as yet, we have not defined a governing equation for the propagation of this interface motion into the fluid domain. For a given fluid domain displacement $\boldsymbol{d}^\Gamma$, the fluid domain deformation gradient is defined as $\boldsymbol{F}^\Gamma = \mathbf{I} + (\boldsymbol{\nabla}\boldsymbol{d}^\Gamma)^{\mathrm{T}}$ and its determinant as $J^\Gamma = \det[\boldsymbol{F}^\Gamma]$.

Employing $\boldsymbol{F}_\Gamma$ and J_Γ, volume continuity can be expressed in terms of an integral over the initial fluid configuration as

$$\oint_{\Gamma_0^f} \left(J\boldsymbol{F}^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right) \cdot \boldsymbol{v}^f \, d\Gamma_0^f = 0 \tag{10}$$

Before expressing linear momentum conservation over the initial configuration, we must introduce the mapping of the gradient operator in the deformed configuration to the gradient operator in the initial configuration:

$$\boldsymbol{\nabla}\boldsymbol{\phi} = \boldsymbol{F}^{-T} \cdot (\boldsymbol{\nabla}_0 \boldsymbol{\phi}) \tag{11}$$

$$(\boldsymbol{\nabla}\boldsymbol{\phi})^T = (\boldsymbol{\nabla}_0 \boldsymbol{\phi})^T \cdot \boldsymbol{F}^{-1} \tag{12}$$

With these mappings, linear momentum conservation for the fluid can be expressed over the initial configuration as

$$\begin{aligned}
\int_{\Omega_0^f} J^\Gamma \frac{\partial \boldsymbol{v}^f}{\partial t}\, d\Omega_0^f + \oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f \right] \cdot \left(\boldsymbol{v}^f - \frac{\partial \boldsymbol{d}^\Gamma}{\partial t} \right) \boldsymbol{v}^f \, d\Gamma_0^f = \\
\oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f \right] \cdot \left[\nu (\boldsymbol{F}^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 \boldsymbol{v}^f) \right] \, d\Gamma_0^f \\
+ \int_{\Omega_0^f} J^\Gamma (\boldsymbol{F}^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 p)\, d\Omega_0^f \quad + \quad \int_{\Omega_0^f} \boldsymbol{f}_{b_0}^f \, d\Omega_0^f
\end{aligned} \tag{13}$$

where the time derivative in the first term is brought inside the integral as the initial volume is unchanging. In addition, the identity $\boldsymbol{\nabla} \cdot \left[\left(\boldsymbol{\nabla} \boldsymbol{v}^f \right)^T \right] = \boldsymbol{\nabla} \left(\boldsymbol{\nabla} \cdot \boldsymbol{v}^f \right) = \boldsymbol{0}$ has been employed, and the velocity $\boldsymbol{v}^\Gamma$ of the moving fluid domain has been replaced with the time derivative $\partial \boldsymbol{d}^\Gamma / \partial t$ of the moving fluid domain displacement.

For the fluid domain, four boundary condition types are considered: inlet, outlet, no-slip wall, and symmetry.

2.3 Fluid-Solid Interface Conditions

At the interface Γ_{fsi} between the solid and fluid regions, mass conservation leads to the *kinematic* interface conditions, which state that the velocity of the fluid is equal to the velocity of the solid and the velocity of the moving fluid domain:

$$\boldsymbol{v}^f = \frac{\partial \boldsymbol{d}^s}{\partial t} = \frac{\partial \boldsymbol{d}^\Gamma}{\partial t} \qquad \text{on} \quad \Gamma_{\text{fsi}} \tag{14}$$

ensuring the fluid and solid share the same motion at the interface (i.e., there is no gap or overlap), assuming the fluid and solid interfaces are coincident in the initial configuration.

Linear momentum conservation yields the *kinetic* interface condition in the deformed configuration Γ_{fsi} as:

$$\hat{\boldsymbol{n}}^f \cdot \boldsymbol{\sigma}^f = -\hat{\boldsymbol{n}}^s \cdot \boldsymbol{\sigma}^s \tag{15}$$

$$\hat{\boldsymbol{n}}^f \cdot \left[\nu \boldsymbol{\nabla} \boldsymbol{v}^f + \nu \left(\boldsymbol{\nabla} \boldsymbol{v}^f \right)^T - p \right] = -\hat{\boldsymbol{n}}^s \cdot \boldsymbol{\sigma}^s \tag{16}$$

where $\hat{\boldsymbol{n}}^f = -\hat{\boldsymbol{n}}^s$, as the surface normals are assumed to point outwards from each region. Equivalently, the kinetic condition can be expressed on the initial configuration as

$$\left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f \right] \cdot \boldsymbol{\sigma}^f = - \left[J \boldsymbol{F}^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^s \right] \cdot \boldsymbol{\sigma}^s \tag{17}$$

$$\left[J^\Gamma(\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right] \cdot \left[\nu(\boldsymbol{F}^\Gamma)^{-T} \cdot \boldsymbol{\nabla} \boldsymbol{v}^f + \nu \left(\boldsymbol{\nabla} \boldsymbol{v}^f\right)^T \cdot (\boldsymbol{F}^\Gamma)^{-1} - p\right] = -\left[J\boldsymbol{F}^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^s\right] \cdot \boldsymbol{\sigma}^s \quad (18)$$

2.4 Fluid Domain Motion

The motion of the fluid-solid interface Γ_{fsi} is dictated by the motion of the solid domain; however, as yet, we have not defined how this interface motion should be distributed throughout the fluid domain. This fluid domain motion need not follow a conservation law and should instead be ideally defined to smoothly transit the interface motion, preserving the quality of the computational mesh—to be described below. Nonetheless, any chosen fluid domain motion procedure or equation should follow the geometric (space) conservation law (Equation 7), which states that the rate of change of the fluid domain volume (or a subset of it) must equal the net flux of the domain velocity across its boundaries.

In the current work, the displacement $\boldsymbol{d}^\Gamma$ of the fluid domain is assumed to follow the pseudo-solid linear elastic, static momentum equation:

$$\oint_{\Gamma_0^f} \boldsymbol{n}_0^f \cdot \left[\mu^\Gamma \boldsymbol{\nabla}_0 \boldsymbol{d}^\Gamma + \mu^\Gamma \left(\boldsymbol{\nabla}_0 \boldsymbol{d}^\Gamma\right)^T + \lambda^\Gamma (\boldsymbol{\nabla}_0 \cdot \boldsymbol{d}^\Gamma)\mathbf{I}\right] d\Gamma_0^f = \mathbf{0} \quad (19)$$

where μ^Γ and λ^Γ are spatially-varying *stiffness* parameters for controlling the distribution of the motion. Note that the quantities are not mapped to the deformed configuration, resulting in the equation being linear in $\boldsymbol{d}^\Gamma$; as a result, this approach does not formally conserve linear momentum, however, as will be shown in test cases section below, it is sufficient for maintaining the fluid domain computational mesh quality. The value of $\boldsymbol{d}^\Gamma$ is prescribed on all boundaries of Γ^f, where the $\boldsymbol{d}^\Gamma = \boldsymbol{d}^s$ on the fluid-solid interface and zero on other boundaries.

3 Numerical Methods

3.1 Spatio-temporal Discretisation

The solution domain is discretised in both space and time. The total simulation period is divided into a finite number of time increments, denoted as Δt, and the discretised governing equations are solved iteratively in a time-marching fashion. The fluid and solid spatial domains are partitioned into a finite set $\mathcal{P}$ of contiguous convex polyhedral cells, where each cell is denoted by $P \in \mathcal{P}$. The total number of cells in the mesh is indicated by $|\mathcal{P}| = |\mathcal{P}|^f + |\mathcal{P}^s|$, where $|\mathcal{P}^f|$ indicates the number of cells in the fluid mesh and $|\mathcal{P}^s|$ indicates the number in the solid mesh. A representative cell P is shown in Figure 2. The set of all faces of cell P is denoted by $\mathcal{F}_P$. This set is further subdivided into three disjoint subsets:

- **Internal Faces** ($\mathcal{F}_P^{\text{int}}$): Faces that are shared with neighbouring cells, excluding the fluid-solid interface.
- **Boundary Faces** ($\mathcal{F}_P^{\text{bnd}}$): Faces that lie on the boundary of the spatial domain, excluding the fluid-solid interface.

- **Fluid-Solid Interface Faces** ($\mathcal{F}_P^{\text{fsi}}$): Faces that lie on the fluid-solid interface; that is, faces that straddle a cell in the fluid mesh and a cell in the solid mesh.

I need to update below: notation becomes too verbose when FSI is added Maybe include as internal face and introduce notation for subsets Each internal face $f_i \in \mathcal{F}_P^{\text{int}}$ corresponds to a neighbouring cell $N_{f_i} \in \mathcal{N}_P$, where $\mathcal{N}_P$ is the set of all neighbouring cells of P in the same region. The outward unit normal vector associated with an internal face f_i is denoted by $\mathbf{n}_{f_i}$, while $\mathbf{n}_{b_i}$ indicates the normal for a boundary face b_i, and $\mathbf{n}_{I_i}$ indicates the normal for a fluid-solid interface face I_i. The vector $\mathbf{d}_{f_i}$ connects the centroid of cell P with the centroid of the neighbouring cell N_{f_i}, whereas the vector $\mathbf{d}_{b_i}$ connects the centroid of cell P with the centroid of boundary face b_i, and the vector $\mathbf{d}_{I_i}$ connects the centroid of cell P with the centroid of the neighbouring cell N_{I_i} across a fluid-solid interface.

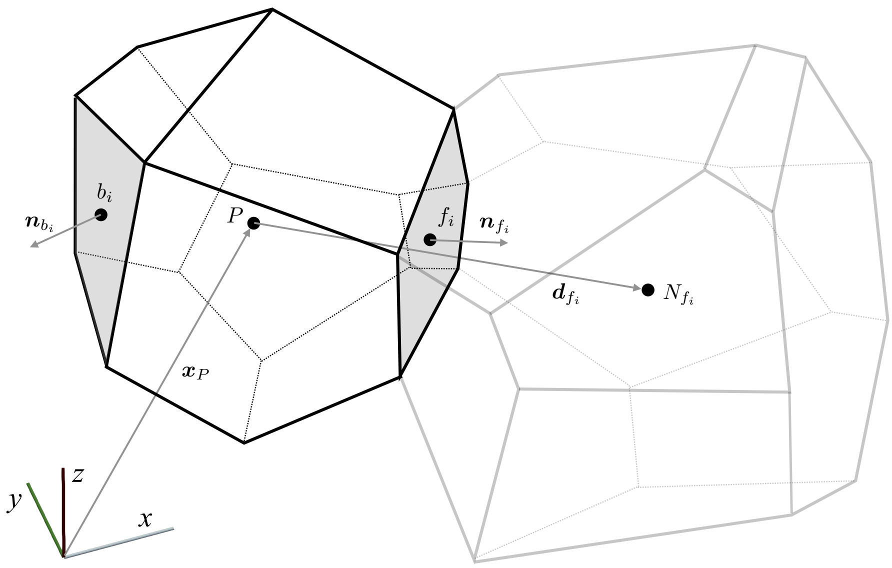


Fig. 2 Representative convex polyhedral cell P and neighbouring cell N_{f_i}, which share a face f_i (Taken from **?**])

3.2 Newton-Type Solution Method

To facilitate the application of a Newton-type solution algorithm, the governing equations are expressed in *residual* form as:

$$\boldsymbol{R}(\boldsymbol{U}) = \begin{pmatrix} \boldsymbol{R}_u \\ R_p \\ \boldsymbol{R}_m \\ \boldsymbol{R}_d \end{pmatrix} = \boldsymbol{0} \tag{20}$$

where $\boldsymbol{R}$ represents the *residual* (imbalance) of the equations, which is a function of the primary unknowns field $\boldsymbol{U} = \{\boldsymbol{v}, p, \boldsymbol{d}_m, \boldsymbol{d}\}$. Concretely, the components of the residual are given as use on

Omega f notation etc

$$
\begin{aligned}
\boldsymbol{R}_u \;=\;& \oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right] \cdot \left[\nu (\boldsymbol{F}^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 \boldsymbol{v}^f)\right] \, d\Gamma_0^f + \int_{\Omega_0^f} J^\Gamma (\boldsymbol{F}^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 p) \, d\Omega_0^f \\
& + \int_{\Omega_0^f} \boldsymbol{f}_{b_0}^f \, d\Omega_0^f - \int_{\Omega_0^f} J^\Gamma \frac{\partial \boldsymbol{v}^f}{\partial t} \, d\Omega_0^f - \oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right] \cdot \left(\boldsymbol{v}^f - \frac{\partial \boldsymbol{d}_m}{\partial t}\right) \boldsymbol{v}^f \, d\Gamma_0^f \\
& + \boldsymbol{\mathcal{D}}_v \\
R_p \;=\;& \oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right] \cdot \boldsymbol{v}^f \, d\Gamma_0^f \;+\; \mathcal{D}_p \\
\boldsymbol{R}_m \;=\;& \oint_{\Gamma_0^f} \boldsymbol{n}_0^f \cdot \left[\mu^\Gamma \boldsymbol{\nabla}_0 \boldsymbol{d}_m + \mu^\Gamma \left(\boldsymbol{\nabla}_0 \boldsymbol{d}_m\right)^T + \lambda^\Gamma (\boldsymbol{\nabla}_0 \cdot \boldsymbol{d}_m) \mathbf{I}\right] d\Gamma_0^f \;+\; \boldsymbol{\mathcal{D}}_m \\
\boldsymbol{R}_d \;=\;& \oint_{\Gamma_0^s} \left(J \boldsymbol{F}^{-\mathrm{T}} \cdot \boldsymbol{n}_0^s\right) \cdot \boldsymbol{\sigma}^s \, d\Gamma_0^s + \int_{\Omega_0^s} \boldsymbol{f}_{b_0}^s \, d\Omega_0^s - \int_{\Omega_0^s} \rho_0 \frac{\partial^2 \boldsymbol{d}}{\partial t^2} \, d\Omega_0^s \;+\; \boldsymbol{\mathcal{D}}_d
\end{aligned} \tag{21}
$$

where $\boldsymbol{\mathcal{D}}_v$, $\mathcal{D}_p$, $\boldsymbol{\mathcal{D}}_m$, and $\boldsymbol{\mathcal{D}}_d$ are stabilisation terms to be defined.

The residuals $\boldsymbol{R}_v$, R_p and $\boldsymbol{R}_m$ are applied to each cell in the fluid domain and discretised, while residual $\boldsymbol{R}_d$ is applied to each cell in the solid domain and discretised, as is described below. The resulting system of coupled nonlinear equations in terms of $\boldsymbol{U}$ can be linearised and iteratively solved using a Newton-type method:

$$
\begin{aligned}
\boldsymbol{J}(\boldsymbol{U}_k)\delta \boldsymbol{U} \;&=\; -\boldsymbol{R}(\boldsymbol{U}_k), \\
\boldsymbol{U}_{k+1} \;&=\; \boldsymbol{U}_k + s\, \delta \boldsymbol{U}, \\
k \;&=\; 0, 1, \ldots
\end{aligned} \tag{22}
$$

where $\boldsymbol{J} \equiv \boldsymbol{R}' \equiv \partial \boldsymbol{R} / \partial \boldsymbol{U}$ is the Jacobian matrix. Starting the Newton procedure requires the specification of $\boldsymbol{U}_0$. The scalar $s > 0$ can be chosen to improve convergence, for example, using a line search or under-relaxation/damping procedure, and is equal to unity in the classic Newton-Raphson approach. Iterations are performed over this system until the residual $\boldsymbol{R}(\boldsymbol{U}_k)$ and solution correction $\delta \boldsymbol{U}$ are sufficiently small, with appropriate normalisation.

The linear system in Equation 22 can be represented as

$$
\begin{bmatrix}
\boldsymbol{J}_{vv} & \boldsymbol{J}_{vp} & \boldsymbol{J}_{vm} & 0 \\
J_{pv} & J_{pp} & J_{pm} & 0 \\
0 & 0 & \boldsymbol{J}_{mm} & \boldsymbol{J}_{md} \\
\boldsymbol{J}_{dv} & \boldsymbol{J}_{dp} & 0 & \boldsymbol{J}_{dd}
\end{bmatrix}_k
\begin{bmatrix}
\delta \boldsymbol{v} \\ \delta p \\ \delta \boldsymbol{d}_m \\ \delta \boldsymbol{d}
\end{bmatrix}
= -
\begin{bmatrix}
\boldsymbol{R}_v \\ R_p \\ \boldsymbol{R}_m \\ \boldsymbol{R}_d
\end{bmatrix}_k \tag{23}
$$

where the subscript k indicates that the Jacobian matrix and right-hand side vector are evaluated in terms of $\boldsymbol{U}_k$. Jvd and Jpd can be removed since this comes form Jvm and Jpm The submatrices $\boldsymbol{J}_{ab}$ represent the coupling in the residual a coming from the unknown b. Several types of coupling exist between the residual components:

- **Fluid-to-solid**, represented by $\boldsymbol{J}_{dv}$ and $\boldsymbol{J}_{dp}$: the fluid forces are passed to the solid by replacing the solid stress $\boldsymbol{\sigma}^s$ with the fluid stress $\boldsymbol{\sigma}^f$ at the fluid-solid interface in $\boldsymbol{R}_d$.

- Merge with below **Solid-to-fluid**, represented by $\boldsymbol{J}_{vd}$ and $\boldsymbol{J}_{pd}$ not required, per se the solid motion is passed to the fluid equations by replacing the fluid velocity $\boldsymbol{v}^f$ by the solid velocity ${}^{\partial \boldsymbol{d}^s}/_{\partial t}$ at the fluid-solid interface in $\boldsymbol{R}_v$ and R_p.
- **Solid-to-mesh**, represented by $\boldsymbol{J}_{md}$: the solid motion is passed to the fluid mesh by replacing the fluid mesh displacement $\boldsymbol{d}_m$ by the solid displacement $\boldsymbol{d}^s$ at the fluid-solid interface in $\boldsymbol{R}_m$.
- Rename **Mesh-to-fluid**, represented by $\boldsymbol{J}_{vm}$ and $\boldsymbol{J}_{pm}$: the fluid mesh motion velocity ${}^{\partial \boldsymbol{d}_m}/_{\partial t}$ appears in the advection term of $\boldsymbol{R}_v$ and the fluid mesh motion displacement $\boldsymbol{d}_m$ appears in the fluid mesh deformation gradients terms ($\boldsymbol{F}_m$, J_m) of $\boldsymbol{R}_v$ and R_p.

Several submatrices are not present, indicating an absence of coupling:

- $\boldsymbol{J}_{mv}$, $\boldsymbol{J}_{mp}$: the fluid mesh motion $\boldsymbol{d}_m$ does not depend on the fluid velocity $\boldsymbol{v}$ and pressure p.
- $\boldsymbol{J}_{dm}$: the solid motion $\boldsymbol{d}$ does not depend on the fluid mesh motion $\boldsymbol{d}_m$.

The current work adopts a Jacobian-free Newton-Krylov solution procedure [**?**] to solve the nonlinear system given by Equation 20. Consequently, the true Jacobian given in Equation 23 is not strictly required; instead, an approximate Jacobian $\tilde{\boldsymbol{J}}$ is formed and employed to precondition the nonlinear Newton-Krylov solution. Note that adopting an approximate Jacobian does not affect the final converged solution in each time step, assuming convergence is achieved; in addition, the quadratic convergence of the Newton-Raphson method is unaffected, once again assuming convergence is achieved. Instead, this approximate Jacobian only affects the convergence of the linear iterative (Krylov) solver. Others have proposed the use of physics-specific preconditioning procedures, e.g. **?**] and **?**] =¿ these approaches sound better, but here we use a simpler approach to demonstrate the first FVM monolithic FSI solver. Physics preconditioners are certainly suitable (e.g., using SIMPLE and segregated solvers) and will be examined in a future publication. Here, the approximate Jacobian $\tilde{\boldsymbol{J}}$ takes the form:

$$\tilde{\boldsymbol{J}} = \begin{bmatrix} \tilde{\boldsymbol{J}}_{vv} & \boldsymbol{J}_{vp} & \tilde{\boldsymbol{J}}_{vm} & 0 \\ \boldsymbol{J}_{pv} & \tilde{J}_{pp} & 0 & 0 \\ 0 & 0 & \tilde{\boldsymbol{J}}_{mm} & \tilde{\boldsymbol{J}}_{md} \\ 0 & \boldsymbol{J}_{dp} & 0 & \tilde{\boldsymbol{J}}_{dd} \end{bmatrix}_k \tag{24}$$

where $\boldsymbol{J}_{dv}$ is dropped as the fluid viscous forces on the fluid-solid interface are assumed to be small in comparison to the fluid pressure forces (represented by $\boldsymbol{J}_{dp}$). In addition, compact stencil approximate Jacobians for solid ($\tilde{\boldsymbol{J}}_{dd}$), fluid ($\tilde{\boldsymbol{J}}_{vv}$, $\tilde{J}_{pp}$) and fluid mesh motion ($\tilde{\boldsymbol{J}}_{mm}$, $\tilde{\boldsymbol{J}}_{md}$) terms are employed. The J_{pm} term is also ignored, and $\tilde{\boldsymbol{J}}_{vm}$ contains only the advective mesh velocity term.

Details of the cell-centred finite volume discretisation of residuals are given below, in addition to calculating the approximate Jacobian submatrices.

3.3 Cell-Centred Finite Volume Discretisation Principles

This section summaries the main principles of the cell-centred FV scheme employed here State truncated Taylor series A nominally second-order cell-centred finite volume discretisation is employed in this work. This section describes the adopted finite volume discretisation for general volume and surface integrals, gradient calculation, and time derivatives. Equation-specific details are given in the subsequent fluid dynamics, mesh motion and solid dynamics sections.

The presented discretisation is second-order accurate in space for displacement if the cell-centred displacement gradients (and the stress) are at least first-order accurate in space, even if the cell faces are not flat.

To quell zero-energy solution modes (i.e. checkerboarding oscillations), Rhie-Chow-type stabilisation terms [6] are added to the residual equations, as described below. As shown in **?**] and **?**], a Rhie-Chow-type stabilisation term can be cast in the same form as an upwind stabilisation term.

Volume Integrals

Volume integrals over a cell P are approximated to second-order accuracy by assuming the integrand $\boldsymbol{\phi}$ to vary locally according to a truncated Taylor series expansion about the centroid:

$$\begin{aligned}\int_{\Omega_P} \boldsymbol{\phi}\, d\Omega_P &\approx \int_{\Omega_P} \left[\boldsymbol{\phi}_P + (\boldsymbol{x} - \boldsymbol{x}_P) \cdot \nabla \boldsymbol{\phi}_P\right] d\Omega_P \\ &\approx \boldsymbol{\phi}_P \Omega_P \end{aligned} \tag{25}$$

where subscript P indicates a value at the centroid of the cell P, Ω_P is the volume of cell P and $\int_{\Omega_P} (\boldsymbol{x} - \boldsymbol{x}_P) d\Omega_P \equiv 0$ by definition of the cell centroid. This approximation corresponds to the midpoint rule and one-point quadrature.

Surface Integrals

Surface integrals of generic integrand $\boldsymbol{q}$ can be discretised using one-point quadrature at each face as

$$\begin{aligned}\oint_{\Gamma_P} \boldsymbol{n} \cdot \boldsymbol{q}\, d\Gamma_P &= \sum_{f_i \in \mathcal{F}_P} \int_{\Gamma_{f_i}} \boldsymbol{n} \cdot \boldsymbol{q}\, d\Gamma_{f_i} \\ &\approx \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \boldsymbol{\Gamma}_{f_i} \cdot \boldsymbol{q}_{f_i} + \sum_{d_i \in \mathcal{F}_P^{\text{D}}} \boldsymbol{\Gamma}_{d_i} \cdot \boldsymbol{q}_P + \sum_{s_i \in \mathcal{F}_P^{\text{symm}}} \boldsymbol{\Gamma}_{s_i} \cdot \boldsymbol{q}_{s_i} + \sum_{t_i \in \mathcal{F}_P^{\text{trac}}} |\boldsymbol{\Gamma}_{t_i}| \bar{\boldsymbol{Q}}_{t_i} \end{aligned} \tag{26}$$

where Γ_P indicates the surface of cell P, $\boldsymbol{\Gamma}_{f_i}$ indicates the deformed area vector of face f_i, and $\bar{\boldsymbol{Q}}_{t_i}$ represents the prescribed normal component of $\boldsymbol{q}$ on a boundary faces where $\boldsymbol{q}$ is prescribed, e.g. $\bar{\boldsymbol{Q}}_{t_i}$ represents the prescribed traction on fluid and solid Neumann boundaries.

In the current work, a unique definition of solution gradient (e.g., $\boldsymbol{q}_{f_i}$) In the current work, the value of the integrand $\boldsymbol{q}$ at each cell face f_i is given as a weighted-averaged of values in the two cells straddling the face [2**?**]:

$$\boldsymbol{q}_{f_i} = w_{f_i} \boldsymbol{q}_P + (1 - w_{f_i}) \boldsymbol{q}_{N_{f_i}} \tag{27}$$

$$\boldsymbol{q}_{f_i} = (1/2) \left[\boldsymbol{q}_P + \boldsymbol{q}_{N_{f_i}}\right] \tag{28}$$

$$w_{f_i} = (\boldsymbol{n}_{f_i} \cdot [\boldsymbol{x}_{N_{f_i}} - \boldsymbol{x}_{f_i}])/(\boldsymbol{n}_{f_i} \cdot [\boldsymbol{x}_{N_{f_i}} - \boldsymbol{x}_P]) \tag{29}$$

where w_{f_i} is the interpolation weight a face f_i. We can average instead for derivative terms - 2nd eq Should I use deformed config weights? Mat props harmonic interp?

The integrand value $\boldsymbol{q}_{s_i}$ at a symmetry boundary face is calculated as

$$\begin{aligned} \boldsymbol{q}_{s_i} &= (1/2)\left(\boldsymbol{q}_P + \boldsymbol{R}_{s_i} \cdot \boldsymbol{q}_P\right) \\ &= \left(\mathbf{I} - \boldsymbol{n}_{s_i} \otimes \boldsymbol{n}_{s_i}\right) \cdot \boldsymbol{q}_P \end{aligned} \tag{30}$$

where $\boldsymbol{R}_{s_i} \cdot \boldsymbol{q}_P$ represents the mirror reflection of $\boldsymbol{q}_P$ across the symmetry boundary face s_i. The reflection tensor is $\boldsymbol{R}_{s_i} = \mathbf{I} - 2\boldsymbol{n}_{s_i} \otimes \boldsymbol{n}_{s_i}$ [3], with $\boldsymbol{n}_{s_i}$ indicating the unit normal of the symmetry boundary face s_i.

The discretisation of surface integral terms above (Equation 26) is given in terms of deformed configuration; integration over the initial domain is achieved by expressing the deformed area vectors $\boldsymbol{\Gamma}_{f_i}$ at a face in terms of the initial configuration:

$$\boldsymbol{\Gamma}_{f_i} = J_{f_i} \boldsymbol{F}_{f_i}^{-\mathrm{T}} \cdot \boldsymbol{\Gamma}_{0_{f_i}} \tag{31}$$

where the face values J_{f_i} and $\boldsymbol{F}_{f_i}^{-\mathrm{T}}$ are averaged from adjacent cells according to Equation 27. Use text(T) for all transposes

Gradients

Cell-centred displacement gradients $\boldsymbol{\nabla}\boldsymbol{\phi}$ are determined using a weighted first-neighbours least squares method [2?]: introduce symbol for LS vec

$$\begin{aligned} (\boldsymbol{\nabla}\boldsymbol{\phi})_P = &\sum_{f_i \in \mathcal{F}_P^{\text{int}}} w_{f_i} |\boldsymbol{\Gamma}_{f_i}| \frac{\boldsymbol{G}_P^{-1} \cdot \boldsymbol{d}_{f_i}}{\boldsymbol{d}_{f_i} \cdot \boldsymbol{d}_{f_i}} \otimes \left(\boldsymbol{\phi}_{N_{f_i}} - \boldsymbol{\phi}_P\right) \\ &+ \sum_{d_i \in \mathcal{F}_P^{\text{D}}} w_{f_i} |\boldsymbol{\Gamma}_{d_i}| \frac{\boldsymbol{G}_P^{-1} \cdot \boldsymbol{d}_{d_i}}{\boldsymbol{d}_{d_i} \cdot \boldsymbol{d}_{d_i}} \otimes \left(\boldsymbol{\phi}_{d_i} - \boldsymbol{\phi}_P\right) \\ &+ \sum_{s_i \in \mathcal{F}_P^{\text{symm}}} w_{f_i} |\boldsymbol{\Gamma}_{s_i}| \frac{\boldsymbol{G}_P^{-1} \cdot \boldsymbol{d}_{s_i}}{\boldsymbol{d}_{s_i} \cdot \boldsymbol{d}_{s_i}} \otimes \left(\boldsymbol{R}_{s_i} \cdot \boldsymbol{\phi}_P - \boldsymbol{\phi}_P\right) \end{aligned} \tag{32}$$

which is exact for linear functions. The tensor multiplication operator $\otimes$ is replaced with an appropriate vector-scalar multiplication operator when $\boldsymbol{\phi}$ is a scalar, e.g. kinematic pressure p. The vector $\boldsymbol{\phi}_{b_i}$ indicates the value of $\boldsymbol{\phi}$ at the centroid of boundary face b_i, while vector $\boldsymbol{d}_{d_i}$ connects the centroid of cell P to the centroid of displacement boundary face d_i. The quantity $\boldsymbol{R}_{s_i} \cdot \boldsymbol{\phi}_P$ represents the mirror reflection of $\boldsymbol{\phi}_P$ across the symmetry boundary face s_i. The vector $\boldsymbol{d}_{s_i}$ connects the centroid $\boldsymbol{x}_P$ of cell P with its mirror reflection $\boldsymbol{R}_{s_i} \cdot \boldsymbol{x}_P$ through boundary face s_i.

The $\boldsymbol{G}_P$ tensor for cell P is calculated as

$$\begin{aligned}\boldsymbol{G}_P = & \sum_{f_i \in \mathcal{F}_P^{\text{int}}} (1 - w_{f_i}) |\boldsymbol{\Gamma}_{f_i}| \frac{\boldsymbol{d}_{f_i} \otimes \boldsymbol{d}_{f_i}}{\boldsymbol{d}_{f_i} \cdot \boldsymbol{d}_{f_i}} \\ & + \sum_{d_i \in \mathcal{F}_P^{\text{disp}}} (1 - w_{d_i}) |\boldsymbol{\Gamma}_{d_i}| \frac{\boldsymbol{d}_{d_i} \otimes \boldsymbol{d}_{d_i}}{\boldsymbol{d}_{d_i} \cdot \boldsymbol{d}_{d_i}} \\ & + \sum_{s_i \in \mathcal{F}_P^{\text{symm}}} (1 - w_{s_i}) |\boldsymbol{\Gamma}_{s_i}| \frac{\boldsymbol{d}_{s_i} \otimes \boldsymbol{d}_{s_i}}{\boldsymbol{d}_{s_i} \cdot \boldsymbol{d}_{s_i}} \end{aligned} \tag{33}$$

As the $w|\boldsymbol{\Gamma}|\ \boldsymbol{G}_P^{-1} \cdot \boldsymbol{d}/(\boldsymbol{d} \cdot \boldsymbol{d})$ vectors in Equation 32 are purely a function of the mesh, they can be computed once (or each time the mesh moves) and stored. Equation 32 approximates the cell-centre gradients to at least a first-order accuracy, increasing to second-order accuracy on certain smooth grids [5]; first-order accurate gradients are sufficient to preserve second-order accuracy of the cell-centre primary unknowns.

Time Derivatives

Time derivatives $\partial \boldsymbol{\phi}/\partial t$ at the centre of cell P are discretised in time using the second-order backwards (BDF2) scheme:

$$\left(\frac{\partial \boldsymbol{\phi}}{\partial t} \right)_P^{[t+1]} \approx \frac{3\boldsymbol{\phi}_P^{[t+1]} - 4\boldsymbol{\phi}_P^{[t]} + \boldsymbol{\phi}_P^{[t-1]}}{2\Delta t} \tag{34}$$

where Δt is the time increment – assumed constant here. Superscript $[t]$ indicates the time level, with $\boldsymbol{\phi}_P^{[t+1]}$ corresponding to the value of $\boldsymbol{\phi}$ at the centre of cell P at the current time step $[t+1]$. Using the second-order backwards scheme, second time derivatives $\partial^2 \boldsymbol{\phi}/\partial t^2$ are discretised as

$$\left(\frac{\partial^2 \boldsymbol{\phi}}{\partial t^2} \right)_P \approx \frac{3 \left(\dfrac{3\boldsymbol{\phi}_P^{[t+1]} - 4\boldsymbol{\phi}_P^{[t]} + \boldsymbol{\phi}_P^{[t-1]}}{2\Delta t} \right) - 4 \left(\frac{\partial \boldsymbol{\phi}}{\partial t} \right)_P^{[t]} + \left(\frac{\partial \boldsymbol{\phi}}{\partial t} \right)_P^{[t-1]}}{2\Delta t} \tag{35}$$

where the $\frac{\partial \boldsymbol{\phi}}{\partial t}$ terms are discretised are also discretised using the second-order backwards scheme (Equation 34).

3.4 Discretisation of the Residuals and Jacobian

3.4.1 Fluid Dynamics Discretisation

This section describes the discretisation of the fluid residuals $\boldsymbol{R}_v$ and R_p, as well as the defining the adopted approximate Jacobian submatrices $\boldsymbol{J}_{vv}$, $\boldsymbol{J}_{vp}$, J_{pp} and J_{pv}.

The fluid equation linear momentum residual $\boldsymbol{R}_v$ repeated here is

$$\begin{aligned}\boldsymbol{R}_u = &\oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right] \cdot \left[\nu (\boldsymbol{F}^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 \boldsymbol{v}^f)\right] d\Gamma_0^f + \int_{\Omega_0^f} J^\Gamma (\boldsymbol{F}^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 p)\, d\Omega_0^f \\ &+ \int_{\Omega_0^f} \boldsymbol{f}_{b_0}^f \, d\Omega_0^f - \int_{\Omega_0^f} J^\Gamma \frac{\partial \boldsymbol{v}^f}{\partial t} d\Omega_0^f - \oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right] \cdot \left(\boldsymbol{v}^f - \frac{\partial \boldsymbol{d}_m}{\partial t}\right) \boldsymbol{v}^f \, d\Gamma_0^f \end{aligned} \tag{36}$$

The three volume integrals are approximated to second-order accuracy using the mid-point rule (Equation 25):

$$\int_{\Omega_0^f} J^\Gamma (\boldsymbol{F}^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 p)\, d\Omega_0^f \approx J_P^\Gamma (\boldsymbol{F}_P^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 p)_P \, \Omega_{0,P} \tag{37}$$

$$\int_{\Omega_0^f} \boldsymbol{f}_{b_0}^f \, d\Omega_0^f \approx \left(\boldsymbol{f}_{b_0}^f\right)_P \Omega_{0,P} \tag{38}$$

$$\int_{\Omega_0^f} J^\Gamma \frac{\partial \boldsymbol{v}^f}{\partial t} d\Omega_0^f \approx J_P^\Gamma \frac{3\boldsymbol{v}_P - 4\boldsymbol{v}_P^{[t]} + \boldsymbol{v}_P^{[t-1]}}{2\Delta t} \Omega_{0,P} \tag{39}$$

where the second-order in time backwards differencing scheme is used for the time derivative; the $[t{+}1]$ time index notation is dropped for brevity, i.e., $\boldsymbol{v}_P \equiv \boldsymbol{v}_P^{[t+1]}$. The cell-centred pressure gradient $(\boldsymbol{\nabla}_0 p)_P$ is calculated using weighted least squares (Equation **??**). The cell-centred fluid mesh deformation gradient and its determinant are calculated in terms of the fluid mesh displacement gradient according to Equation **??**, where the cell-centred gradient is calculated using weighted least squares (Equation **??**).

The diffusion (viscous stresses) surface integral is approximated as FSI-term

$$\begin{aligned}&\oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right] \cdot \left[\nu (\boldsymbol{F}^\Gamma)^{-T} \cdot (\boldsymbol{\nabla}_0 \boldsymbol{v}^f)\right] d\Gamma_0^f \approx \\ &\qquad \sum_{f_i \in \mathcal{F}_P^{\mathrm{int}}} \left[J_{f_i}^\Gamma (\boldsymbol{F}_{f_i}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{\Gamma}}_{0,f_i}^f\right] \cdot \left[\nu_{f_i} (\boldsymbol{F}_{f_i}^\Gamma)^{-T} \cdot \left(\boldsymbol{\nabla}_0 \boldsymbol{v}^f\right)_{f_i}\right]\end{aligned} \tag{40}$$

The face values $J_{f_i}^\Gamma$, $\boldsymbol{F}_{f_i}^\Gamma$, ν_{f_i}, and $\left(\boldsymbol{\nabla}_0 \boldsymbol{v}^f\right)_{f_i}$ are interpolated from the cell-centres using Equation **??**. Once again, the cell-centred gradient is calculated using weighted least squares (Equation **??**).

The advection surface integral is approximated as FSI-term

$$\begin{aligned}&\oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{n}}_0^f\right] \cdot \left(\boldsymbol{v}^f - \frac{\partial \boldsymbol{d}_m}{\partial t}\right) \boldsymbol{v}^f \, d\Gamma_0^f \approx \\ &\qquad \sum_{f_i \in \mathcal{F}_P^{\mathrm{int}}} \left[J_{f_i}^\Gamma (\boldsymbol{F}_{f_i}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{\Gamma}}_{0,f_i}^f\right] \cdot \left(\boldsymbol{v}_{f_i} - \frac{3\boldsymbol{d}_{m,f_i} - 4\boldsymbol{d}_{m,f_i}^{[t]} + \boldsymbol{d}_{m,f_i}^{[t-1]}}{2\Delta t}\right) \tilde{\boldsymbol{v}}_{f_i} \\ &\qquad + \sum_{d_i \in \mathcal{F}_P^{\mathrm{D}}} \left[J_{d_i}^\Gamma (\boldsymbol{F}_{d_i}^\Gamma)^{-\mathrm{T}} \cdot \hat{\boldsymbol{\Gamma}}_{0,d_i}^f\right] \cdot \left(\boldsymbol{v}_{d_i} - \frac{\partial \boldsymbol{d}_{m,d_i}}{\partial t}\right) \tilde{\boldsymbol{v}}_{d_i}\end{aligned} \tag{41}$$

Note that the advection term is zero at the fluid-solid interface as the face flux (mass flow) is zero by definition. Here, $\tilde{\boldsymbol{v}}_{f_i}$ is calculated using first-order upwind for the approximate Jacobian $\boldsymbol{J}_v$

contribution make consistent with Jacobian section later

$$\tilde{\boldsymbol{v}}_{f_i} = \max\left[\psi_{f_i},\, 0\right] \boldsymbol{v}_P + \min\left[\psi_{f_i},\, 0\right] \boldsymbol{v}_{N_i} \tag{42}$$

where $\max[\cdot\,,\cdot]$ is the maximum operator and $\min[\cdot\,,\cdot]$ is the minimum operator. The volume flux ψ through the control volume face is $\psi = \left(J_m \boldsymbol{F}_m^{-\mathrm{T}} \cdot \boldsymbol{\Gamma}_0\right) \cdot \left(\boldsymbol{v} - \frac{\partial \boldsymbol{d}_m}{\partial t}\right)$. In the residual $\boldsymbol{R}_v$ (which controls the order of accuracy of the final solution), $\tilde{\boldsymbol{v}}_{f_i}$ is calculated using a second-order upwind/CD/limited-decide scheme:

$$\tilde{\boldsymbol{v}}_{f_i} = w_{f_i} \boldsymbol{v}_P + (1 - w_{f_i}) \boldsymbol{v}_{N_{f_i}} \tag{43}$$

The vector stabilisation term $\boldsymbol{\mathcal{D}}_v$ takes the form of the difference between a compact and large stencil gradient at a cell face:

$$\boldsymbol{\mathcal{D}}_v = \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \alpha_v \nu \left[|\boldsymbol{\Delta}_{0,f_i}| \frac{\boldsymbol{v}_{N_{f_i}} - \boldsymbol{v}_P}{|\boldsymbol{\delta}_{0,f_i}|} - \boldsymbol{\Delta}_{0,f_i} \cdot \left(\boldsymbol{\nabla}_0 \boldsymbol{v}\right)_{f_i} \right] |\boldsymbol{\Gamma}_{0,f_i}| \tag{44}$$

where α_v is a user parameter for controlling the amount of stabilisation ($\alpha_v = 0$ gives no stabilisation, $\alpha_v > 0$ gives stabilisation).

The diagonal block (3×3 for 3-D, 2×2 for 2-D) for cell P (row P, column P) is: consistent with JFNK paper, but we can do better by including J/F

$$\begin{aligned}\left\{\tilde{\boldsymbol{J}}_{vv}\right\}_{PP} =& - \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \alpha_v \nu \frac{|\boldsymbol{\Delta}_0|}{|\boldsymbol{\delta}_0|} |\boldsymbol{\Gamma}_0| \, \mathbf{I} \\ & - \frac{3}{2} \frac{J_m \Omega_{0,P}}{\Delta t} \mathbf{I} \quad - \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \max\left[\psi_{f_i}, 0\right] \mathbf{I} + w_{f_i} \left(H_P \boldsymbol{v}_P + H_Q \boldsymbol{v}_Q\right) \otimes \boldsymbol{\Gamma}_{f_i}\end{aligned} \tag{45}$$

where the vectors $\boldsymbol{\Delta}_0 = {\boldsymbol{\delta}_{0,m}}/{\boldsymbol{\delta}_{0,m} \cdot \boldsymbol{n}_{0,m}}$. The Heaviside function terms are defined as

$$\begin{aligned} H_P &= H(\psi_{f_i}) = \begin{cases} 1 & \text{if} \quad \psi_{f_i} > 0 \\ 0 & \text{otherwise} \end{cases} \\ H_Q &= 1 - H(\psi_{f_i}) \end{aligned} \tag{46}$$

while the off-diagonal contribution (row P, column $Q \neq P$) is

$$\left\{\tilde{\boldsymbol{J}}_{vv}\right\}_{PQ} = \nu \frac{|\boldsymbol{\Delta}_0|}{|\boldsymbol{d}_0|} |\boldsymbol{\Gamma}_0| \, \mathbf{I} \quad + \quad \min\left[\psi_{f_i}, 0\right] \mathbf{I} + (1 - w_{f_i}) \left(H_P \boldsymbol{v}_P + H_Q \boldsymbol{v}_Q\right) \otimes \boldsymbol{\Gamma}_{f_i} \tag{47}$$

The block (3×1 for 3-D, 2×1 for 2-D) diagonal coefficient for cell P (row P, column P) for Jacobian submatrix $\boldsymbol{J}_{vp}$ is

$$\left\{\tilde{\boldsymbol{J}}_{vp}\right\}_{PP} = \sum_{f_i \in \mathcal{F}_P^{\text{int}}} J_{P,m} \, \boldsymbol{g}_{P,f_i} \, \Omega_{0,P} \tag{48}$$

where $\boldsymbol{g}_{P,f_i}$ is the least squares vector stored for cell P at face f_i, as defined in Equation ??. While the off-diagonal contribution (row P, column $Q \neq P$) is

$$\left\{ \tilde{\boldsymbol{J}}_{vp} \right\}_{PQ} = -J_{P,m}\, \boldsymbol{g}_{P,f_i}\, \Omega_{0,P} \tag{49}$$

The coupling matrix $\boldsymbol{J}_{vm} \equiv \partial \boldsymbol{R}_v / \partial \boldsymbol{d}_m$ accounts for the effect of the fluid mesh motion on the fluid momentum equation. This effect comes in two three? FSI velocity forms: firstly, the mesh velocity directly appears in the advection term; secondly, all terms contain $\boldsymbol{F}_m$ and J_m, which map the terms from the initial fluid mesh to the unknown deformed mesh. The block (3×3 for 3-D, 2×2 for 2-D) diagonal coefficient for cell P (row P, column P) is expressed as

$$\left[\tilde{\boldsymbol{J}}_{vm} \right]_{PP} = \frac{\alpha_t}{\Delta t} \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \left(J_m \boldsymbol{F}_m^{-\text{T}} \cdot \boldsymbol{\Gamma}_m \right)_{f_i} \otimes \tilde{\boldsymbol{v}}_{f_i} \tag{50}$$

where $\tilde{\boldsymbol{v}}_{f_i}$ is defined in Equation 42. For the first-order Euler scheme $\alpha_t = 1$, while for the second-order backward scheme $\alpha_t = 3/2$. The off-diagonal coefficients (row P, column $Q \neq P$) are

$$\left[\tilde{\boldsymbol{J}}_{vm} \right]_{PQ} = \frac{\alpha_t}{\Delta t} \left(J_m \boldsymbol{F}_m^{-\text{T}} \cdot \boldsymbol{\Gamma}_m \right)_{PQ}^{[m]} \otimes \tilde{\boldsymbol{v}}_{f_i} \tag{51}$$

where subscript PQ indicates a quantity associated with the internal face f shared between cells P and Q.

Moving to the pressure equation, the fluid continuity equation residual R_p repeated here is

$$R_p = \oint_{\Gamma_0^f} \left[J^\Gamma (\boldsymbol{F}^\Gamma)^{-\text{T}} \cdot \hat{\boldsymbol{n}}_0^f \right] \cdot \boldsymbol{v}^f \, d\Gamma_0^f \; + \; \mathcal{D}_p \tag{52}$$

The velocity divergence term is approximated for cell P as

$$\oint_{\Gamma^f} \left[J_m \boldsymbol{F}_m^{-\text{T}} \cdot \hat{\boldsymbol{n}}_0 \right] \cdot \boldsymbol{v} \, d\Gamma_0 \approx \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \left(J_m \boldsymbol{F}_m^{-\text{T}} \cdot \hat{\boldsymbol{\Gamma}}_0 \right) \cdot \left[w_{f_i} \boldsymbol{v}_P + (1 - w_{f_i}) \boldsymbol{v}_{N_{f_i}} \right] \tag{53}$$

where the linear interpolation weights w_{f_i} are given by Equation 29.

The scalar stabilisation term $\mathcal{D}_p$ takes the same form as the vector stabilisation $\mathcal{D}_v$, and is calculated as use Hiro's alpha damping ref vs def config?:

$$\mathcal{D}_p = \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \alpha_p D_p \left[|\boldsymbol{\Delta}_{0,f_i}| \frac{p_{N_{f_i}} - p_P}{|\boldsymbol{\delta}_{0,f_i}|} - \boldsymbol{\Delta}_{0,f_i} \cdot (\boldsymbol{\nabla}_0 p)_{f_i} \right] |\boldsymbol{\Gamma}_{0,f_i}| \tag{54}$$

where, like α_v, α_p is a user parameter for controlling the amount of stabilisation.

The approximate Jacobian submatrix $\tilde{J}_{pp}$ is derived from the compact stencil term of the stabilisation $\mathcal{D}_p$. The block (1×1 for 3-D and 2-D) diagonal coefficient for cell P (row P, column

P) is

$$\left\{\tilde{J}_{pp}\right\}_{PP} = - \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \alpha_p D_p \frac{|\boldsymbol{\Delta}_{0,f_i}|}{|\boldsymbol{\delta}_{0,f_i}|} |\boldsymbol{\Gamma}_{0,f_i}| \tag{55}$$

while the off-diagonal coefficient (row P, column $Q \neq P$) is

$$\left\{\tilde{J}_{pp}\right\}_{PQ} = \alpha_p D_p \frac{|\boldsymbol{\Delta}_{0,f_i}|}{|\boldsymbol{\delta}_{0,f_i}|} |\boldsymbol{\Gamma}_{0,f_i}| \tag{56}$$

For the approximate Jacobian submatrix $\tilde{\boldsymbol{J}}_{pv}$, the block ($1 \times 3$ for 3-D, 1×2 in 2-D) diagonal coefficient for cell P (row P, column P) is

$$\left\{\tilde{\boldsymbol{J}}_{pv}\right\}_{PP} = \sum_{f_i \in \mathcal{F}_P^{\text{int}}} w_{f_i} J_{m,f_i} \boldsymbol{F}_{m,f_i}^{-\text{T}} \cdot \hat{\boldsymbol{\Gamma}}_{0,f_i} \tag{57}$$

while the off-diagonal contribution is

$$\left\{\tilde{\boldsymbol{J}}_{pv}\right\}_{PQ} = (1 - w_{f_i}) J_{m,f_i} \boldsymbol{F}_{m,f_i}^{-\text{T}} \cdot \hat{\boldsymbol{\Gamma}}_{0,f_i} \tag{58}$$

3.4.2 Discretisation of the Solid Dynamics Residual $\boldsymbol{R_d}$

The solid linear momentum equation residual $\boldsymbol{R}_d$ repeated here is

$$\boldsymbol{R}_d = \oint_{\Gamma_0^s} \left(J \boldsymbol{F}^{-\text{T}} \cdot \boldsymbol{n}_0^s\right) \cdot \boldsymbol{\sigma}^s \, d\Gamma_0^s + \int_{\Omega_0^s} \boldsymbol{f}_{b_0}^s \, d\Omega_0^s - \int_{\Omega_0^s} \rho_0 \frac{\partial^2 \boldsymbol{d}}{\partial t^2} \, d\Omega_0^s + \boldsymbol{\mathcal{D}}_d \tag{59}$$

The two volume integrals are approximated to second-order accuracy using the mid-point rule (Equation 25):

$$\int_{\Omega_0^s} \rho_0 \frac{\partial^2 \boldsymbol{d}}{\partial t^2} \, d\Omega_0^s \approx \frac{\rho_{0,P}}{2\Delta t} \left[3 \left(\frac{3\boldsymbol{d}_P - 4\boldsymbol{d}_P^{[t]} + \boldsymbol{d}_P^{[t-1]}}{2\Delta t}\right) - 4\boldsymbol{v}_P^{[t]} + \boldsymbol{v}_P^{[t-1]}\right] \Omega_{0,P} \tag{60}$$

$$\int_{\Omega_0^s} \boldsymbol{f}_{b_0}^s \, d\Omega_0^s \approx \rho_{0,P} \left(\boldsymbol{f}_{b_0}^s\right)_P \Omega_{0,P} \tag{61}$$

where the acceleration term has been approximated using the second-order in time backwards differencing (BDF2) scheme. Here, $\boldsymbol{v} = {}^{\partial \boldsymbol{u}}/_{\partial t}$ refers to the solid velocity vector (not to be confused with the fluid velocity vector), also calculated using the BDF2 scheme:

$$\boldsymbol{v}_P^{[t+1]} \approx \frac{3\boldsymbol{u}_P^{[t+1]} - 4\boldsymbol{u}_P^{[t]} + \boldsymbol{u}_P^{[t-1]}}{2\Delta t} \tag{62}$$

The surface integral term, corresponding to the divergence of stress is discretised as

$$\oint_{\Gamma_0^s} \left(J\boldsymbol{F}^{-\mathrm{T}} \cdot \boldsymbol{n}_0^s\right) \cdot \boldsymbol{\sigma}^s \, d\Gamma_0^s \approx \sum_{f_i \in \mathcal{F}_P^{\mathrm{int}}} \left(J_{f_i} \boldsymbol{F}_{f_i}^{-\mathrm{T}} \cdot \boldsymbol{\Gamma}_{0,f_i}^s\right) \cdot \boldsymbol{\sigma}_{f_i} + \sum_{d_i \in \mathcal{F}_P^{\mathrm{disp}}} \boldsymbol{\Gamma}_{d_i} \cdot \boldsymbol{\sigma}_P + \sum_{t_i \in \mathcal{F}_P^{\mathrm{trac}}} |\boldsymbol{\Gamma}_{t_i}| \bar{\boldsymbol{T}}_{t_i} \tag{63}$$

where vector $\bar{\boldsymbol{T}}_{t_i}$ represents the prescribed traction on the traction boundary face t_i. For faces f_i in Equation 63 that are on traction boundary faces $b_i \in \mathcal{F}_P^{\mathrm{trac}} \subset \mathcal{F}_P$, the traction $\bar{\boldsymbol{t}}_{b_i}$ is known and is directly enforced; that is, $\boldsymbol{\Gamma}_{f_i} \cdot \boldsymbol{\sigma}_{f_i} = |\boldsymbol{\Gamma}_{f_i}| \boldsymbol{n}_{f_i} \cdot \boldsymbol{\sigma}_{f_i} = |\boldsymbol{\Gamma}_{f_i}| \bar{\boldsymbol{t}}_{b_i}$. The stress $\boldsymbol{\sigma}_{f_i}$ at an internal face is calculated by linear interpolation from the adjacent cell centres (Equation **??**). Note from Equation 63 that the stress on displacement boundary faces is assumed to be equal to the stress $\boldsymbol{\sigma}_P$ at the centroid of cell P.

Fluid traction on FSI interface

The cell-centred stress $\boldsymbol{\sigma}_P$ is calculated as a function of the displacement gradient according to the chosen mechanical law, for example, as shown in Appendix **??**.

The stabilisation term $\boldsymbol{\mathcal{D}}_{d,P}$ for a cell P takes the following form two d: new symbol deformed config?:

$$\boldsymbol{\mathcal{D}}_{d,P} = \sum_{f_i \in \mathcal{F}_P^{\mathrm{int}}} \alpha \bar{K}_{f_i} \left[|\boldsymbol{\Delta}_{f_i}| \frac{\boldsymbol{d}_{N_{f_i}} - \boldsymbol{d}_P}{|\boldsymbol{\delta}_{f_i}|} - \boldsymbol{\Delta}_{f_i} \cdot (\boldsymbol{\nabla} \boldsymbol{d})_{f_i} \right] |\boldsymbol{\Gamma}_{f_i}| + \sum_{d_i \in \mathcal{F}_P^{\mathrm{disp}}} \alpha \bar{K}_{d_i} \left[|\boldsymbol{\Delta}_{d_i}| \frac{\bar{\boldsymbol{u}}_{d_i} - \boldsymbol{u}_P}{|\boldsymbol{\delta}_{d_i}|} - \boldsymbol{\Delta}_{d_i} \cdot (\boldsymbol{\nabla} \boldsymbol{d})_P \right] |\boldsymbol{\Gamma}_{d_i}| \tag{64}$$

where $\alpha_d > 0$ is a user-defined parameter for globally scaling the amount of stabilisation, similar to α_v and α_p. Parameter $\bar{K}$ is a stiffness-type parameter that gives the stabilisation an appropriate scale and dimension, chosen here to be $\bar{K} = \frac{4}{3}\mu + \kappa = 2\mu + \lambda$ following previous work [7–9], where μ is the shear modulus (first Lamé parameter), κ is the bulk modulus, and λ is the second Lamé parameter (or their equivalent for the chosen mechanical law). Vector $\bar{\boldsymbol{d}}_{d_i}$ represents the prescribed displacement at the centroid of displacement boundary face d_i. The displacement gradient $(\boldsymbol{\nabla} \boldsymbol{d})_{f_i}$ at the internal face f_i is calculated by interpolation from adjacent cell centres (like in Equation **??**). Note that the displacement gradient at the displacement boundary d_i is assumed to be equal to the displacement gradient at the centroid of cell P. In addition, no stabilisation term is applied on a traction boundary face t_i.

In the current work, an approximate Jacobian $\tilde{\boldsymbol{J}}_{dd}$ is employed for the solid dynamics based on the segregated solution procedures commonly employed in cell-centred finite volume solid dynamics. A recent article [**?**] by the authors describes this approach in detail. The approximate Jacobian $\tilde{\boldsymbol{J}}_{dd}$ is derived from the inertia term and the compact stencil component of the stabilisation term $\boldsymbol{\mathcal{D}}_d$. The $M_d \times M_d$ matrix $\left[\tilde{\boldsymbol{J}}_{dd}\right]$ is symmetric, where $M = 3|\mathcal{P}^s|$ in 3-D and $M = 2|\mathcal{P}^s|$ in 2-D.

The block (3×3 for 3-D, 2×2 for 2-D) diagonal coefficient for cell P (row P, column P) is

$$\left[\tilde{\boldsymbol{J}}_{dd}\right]_{PP} = -\sum_{f_i \in \mathcal{F}_P^{\text{int}}} \alpha \bar{K} \frac{|\boldsymbol{\Delta}_{f_i}|}{|\boldsymbol{\delta}_{f_i}|} |\boldsymbol{\Gamma}_{f_i}| \mathbf{I} \quad - \sum_{d_i \in \mathcal{F}_P^{\text{disp}}} \alpha \bar{K} \frac{|\boldsymbol{\Delta}_{d_i}|}{|\boldsymbol{\delta}_{d_i}|} |\boldsymbol{\Gamma}_{d_i}| \mathbf{I} \quad - \quad \frac{9}{4} \frac{\rho_P \Omega_P}{\Delta t^2} \mathbf{I} \tag{65}$$

while the off-diagonal coefficients (row P, column $Q \neq P$) are

$$\left[\tilde{\boldsymbol{J}}_{dd}\right]_{PQ} = \alpha \bar{K} \frac{\left|\boldsymbol{\Delta}_{f_{PQ}}\right|}{\left|\boldsymbol{d}_{f_{PQ}}\right|} \left|\boldsymbol{\Gamma}_{f_{PQ}}\right| \mathbf{I} \tag{66}$$

where subscript PQ indicates a quantity associated with the internal face f shared between cells P and Q.

The coupling matrix $\boldsymbol{J}_{dp} \equiv \partial \boldsymbol{R}_d / \partial p$ accounts for the effect of the fluid interface pressure on the solid deformation. The block (3×1 for 3-D, 2×1 for 2-D) coefficient for row P, column Q is expressed here as

$$\left[\tilde{\boldsymbol{J}}_{dp}\right]_{PQ} = -\rho^f \left[J^s (\boldsymbol{F}^s)^{-\mathrm{T}} \cdot \hat{\boldsymbol{\Gamma}}_0^s\right] \tag{67}$$

for all pairs of fluid-solid cells at the fluid-solid interface, where P indicates the solid cell index (and row of $\boldsymbol{J}_{dp}$) and Q indicates the fluid cell index (and column of $\boldsymbol{J}_{dp}$). Note that the fluid density ρ^f scales the fluid kinematic pressure to actual pressure, and viscous effects are neglected in the approximate Jacobian (but not in the residual $\boldsymbol{\mathcal{R}}_d$).

3.4.3 Discretisation of the Fluid Mesh Motion Residual $\boldsymbol{R_m}$

The fluid mesh motion equation residual $\boldsymbol{R}_m$ repeated here is maybe we should use large strain approach - see if required in test cases

$$\boldsymbol{R}_m = \oint_{\Gamma_0^f} \boldsymbol{n}_0^f \cdot \left[\mu^\Gamma \boldsymbol{\nabla}_0 \boldsymbol{d}_m + \mu^\Gamma \left(\boldsymbol{\nabla}_0 \boldsymbol{d}_m\right)^T + \lambda^\Gamma (\boldsymbol{\nabla}_0 \cdot \boldsymbol{d}_m) \mathbf{I}\right] d\Gamma_0^f + \boldsymbol{\mathcal{D}}_m \tag{68}$$

The discretisation of the stress divergence is

$$\begin{aligned} &\oint_{\Gamma_0^f} \boldsymbol{n}_0^f \cdot \left[\mu^\Gamma \boldsymbol{\nabla}_0 \boldsymbol{d}_m + \mu^\Gamma \left(\boldsymbol{\nabla}_0 \boldsymbol{d}_m\right)^T + \lambda^\Gamma (\boldsymbol{\nabla}_0 \cdot \boldsymbol{d}_m) \mathbf{I}\right] d\Gamma_0^f \approx \\ &\qquad \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \boldsymbol{\Gamma}_{0,f_i} \cdot \left[\mu_m \left(\boldsymbol{\nabla}_0 \boldsymbol{d}_m\right)_{f_i} + \mu_m \left(\boldsymbol{\nabla}_0 \boldsymbol{d}_m\right)_{f_i}^T + \lambda_m (\boldsymbol{\nabla}_0 \cdot \boldsymbol{d}_m)_{f_i} \mathbf{I}\right] \end{aligned} \tag{69}$$

where the gradient terms are interpolated from the cell centres to the faces according to Equation 27.

The vector stabilisation term $\boldsymbol{\mathcal{D}}_m$ takes the form as $\boldsymbol{\mathcal{D}}_v$:

$$\boldsymbol{\mathcal{D}}_m = \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \alpha_m (2\mu_m + \lambda_m) \left[|\boldsymbol{\Delta}_{0,f_i}| \frac{\boldsymbol{d}_{m,N_{f_i}} - \boldsymbol{d}_{m,P}}{|\boldsymbol{\delta}_{0,f_i}|} - \boldsymbol{\Delta}_{0,f_i} \cdot (\boldsymbol{\nabla}_0 \boldsymbol{d}_m)_{f_i} \right] |\boldsymbol{\Gamma}_{0,f_i}| \tag{70}$$

Merge in text below An additional challenge is that the fluid mesh should be moved (smoothed) to distribute the FSI interface motion throughout the fluid domain. To achieve this, the current work adopts a pseudo-solid approach where the fluid mesh is considered to be a solid domain, where the boundary displacements are prescribed. The following governing equation is adopted for the fluid mesh motion: I could add F-old to rotate but keep linear

$$\oint_{\tilde{\Gamma}} \boldsymbol{n}_u \cdot \left[\beta_m \mu_m \boldsymbol{\nabla} \boldsymbol{d}_m + \beta_m \mu_m (\boldsymbol{\nabla} \boldsymbol{d}_m)^T + \beta_m \lambda_m \text{tr}(\boldsymbol{\nabla} \boldsymbol{d}_m) \mathbf{I} \right] d\tilde{\Gamma} = \mathbf{0} \tag{71}$$

It should be noted that Equation 71 is integrated over the fluid reference configuration, but it is not mapped to the deformed configuration; this choice is made purely from an efficiency perspective: the purpose of this equation is merely to maintain fluid mesh quality, and it need not strictly correspond to a physical system. To maintain fluid mesh quality near the FSI interface and boundary, the stiffness parameters μ_m and λ_m are defined to be inversely proportional to the distance from the fluid boundary according to: express with if-else

$$\beta_m = \min(\beta_{\max} \bar{\delta}^{-1}, \max(\beta_{\min} \bar{\delta}^{-1}, \delta^{-1})) \tag{72}$$

where δ is the (approximate) distance from the fluid boundary. In the current work, δ is calculated using a topological search approach, implemented in the `meshWave` class of OpenFOAM. β_m is then normalised with respect to its average value, i.e. $\beta_m \coloneqq \frac{\beta_m}{\bar{\beta}_m}$, such that $\bar{\beta}_m \equiv 1$. introduce additional terms to avoid confusion

The current work employs a similar approach for defining an approximate Jacobian $\tilde{\boldsymbol{J}}_{mm}$ as is employed for the solid dynamics. based on the segregated solution procedures commonly employed in cell-centred finite volume solid dynamics. The approximate Jacobian $\tilde{\boldsymbol{J}}_{mm}$ is derived from the compact stencil component of the stabilisation term $\boldsymbol{\mathcal{D}}_m$. The $M_m \times M_m$ matrix $\left[\tilde{\boldsymbol{J}}_{mm}\right]$ is symmetric and strongly diagonally dominant (due to Dirichlet conditions), where $M_m = 3|\mathcal{P}^f|$ in 3-D and $M_m = 2|\mathcal{P}^f|$ in 2-D. The block (3×3 for 3-D, 2×2 for 2-D) diagonal coefficient for cell P (row P, column P) can be expressed as notation

$$\left[\tilde{\boldsymbol{J}}_{mm}\right]_{PP} = - \sum_{f_i \in \mathcal{F}_P^{\text{int}}} \alpha \bar{K} \frac{|\boldsymbol{\Delta}_{f_i}|}{|\boldsymbol{\delta}_{f_i}|} |\boldsymbol{\Gamma}_{f_i}| \mathbf{I} \quad - \sum_{d_i \in \mathcal{F}_P^{\text{disp}}} \alpha \bar{K} \frac{|\boldsymbol{\Delta}_{d_i}|}{|\boldsymbol{\delta}_{d_i}|} |\boldsymbol{\Gamma}_{d_i}| \mathbf{I} \tag{73}$$

while the off-diagonal coefficients (row P, column Q) can be expressed as

$$\left[\tilde{\boldsymbol{J}}_{mm}\right]_{PQ} = \alpha \bar{K} \frac{|\boldsymbol{\Delta}_{f_{PQ}}|}{|\boldsymbol{\delta}_{f_{PQ}}|} |\boldsymbol{\Gamma}_{f_{PQ}}| \mathbf{I} \tag{74}$$

where subscript PQ indicates a quantity associated with the internal face f shared between cells P and Q.

The true Jacobian $\boldsymbol{J}_{md} \equiv \partial \boldsymbol{R}_m / \partial \boldsymbol{d}$; in the current work, an approximate coupling matrix $\tilde{\boldsymbol{J}}_{md}$ is employed, where the fluid-solid interface displacement for the fluid mesh interface face is taken as the displacement of the adjacent solid cell centre. This results in a small (one-cell) stencil, in contrast to the true Jacobian $\boldsymbol{J}_{md}$, which would require a larger stencil due to extrapolation from the adjacent solid cell centre using the solid cell centre displacement gradient.

The block (3×3 for 3-D, 2×2 for 2-D) coefficient for row P, column Q can be expressed as ref or current?

$$\left[\tilde{\boldsymbol{J}}_{md}\right]_{PQ} = \beta_m (2\mu_m + \lambda_m) \frac{|\boldsymbol{\Gamma}_{f_i}|}{\boldsymbol{n} \cdot \boldsymbol{d}_{f_i}} \tag{75}$$

for all pairs of fluid-solid cells at the fluid-solid interface, where P indicates the fluid cell index (and row of $\boldsymbol{J}_{md}$) and Q indicates the solid cell index (and column of $\boldsymbol{J}_{md}$).

4 Solution Algorithms

The purpose of preconditioning the Jacobian-free Newton-Krylov method is to reduce the number of inner linear solver iterations. The current work uses the GMRES linear solver for the inner system.

Concretely, the preconditioner needs to approximately solve the linear system $\boldsymbol{y} = \boldsymbol{P}^{-1}\boldsymbol{v}$.

In the current work, the PETSc package (version 3.22.2) [13] is used as the nonlinear solver, driven by a finite volume solver implemented in the solids4foam toolbox [9, 14] for OpenFOAM toolbox [15] (version OpenFOAM-v2312). The codes are publicly available at https://github.com/solids4foam/solids4foam on the `feature-petsc-snes` branch, and the cases and plotting scripts are available at https://github.com/solids4foam/solid-benchmarks.

Globalisation

4.1 Checking Convergence

Update Mention normalisation of solid/mesh residuals The current work adopts three checks for determining whether convergence has been achieved within each time (or loading) step, closely following the default strategy in the PETSc toolbox:

- Norm of the residual $|\boldsymbol{R}(\boldsymbol{u})|$:
  - *Absolute*: Convergence is declared when $|\boldsymbol{R}(\boldsymbol{u})|$ falls below an absolute tolerance a_{tol}, taken here as 1×10^{-50}. This acts as a failsafe in situations where the *relative* criterion (below) is ineffective because the initial solution already nearly satisfies the governing equation.
  - *Relative*: Convergence is declared when $|\boldsymbol{R}(\boldsymbol{u})|$ falls below $r_{\text{tol}} \times |\boldsymbol{R}(\boldsymbol{u}_0)|$, with $|\boldsymbol{R}(\boldsymbol{u}_0)|$ denoting the residual norm at the start of the time (or loading) step. In this work, $r_{\text{tol}} = 10^{-6}$.

- Norm of the solution correction $|\delta \boldsymbol{u}|$: Convergence is declared when the norm of the change in the solution between successive iterations is less than $s_{\text{tol}} \times |\boldsymbol{u}|$.

Although applying the same tolerances for the segregated and Jacobian-free Newton-Krylov solution procedures may seem reasonable, care must be taken with s_{tol}. The Jacobian-free Newton-Krylov method typically makes a small number of large corrections to the solution, whereas the segregated procedure often makes a large number of small corrections. Consequently, if the same s_{tol} value were used, the segregated procedure might be declared convergent prematurely. To avoid this, s_{tol} is set to zero in the present work, meaning convergence is based solely on residual reduction. Nonetheless, setting $s_{\text{tol}} > 0$ could yield a more robust procedure capable of handling potential stalling.

5 Test Cases

5.1 Case 1

WIP

6 Conclusions

WIP The key conclusions of the work are:

- **WIP**: WIP

WIP, yet several areas for future exploration remain:

1. WIP: WIP
2. Higher order discretisation: WIP

Data Availability. CHECK The codes presented are publicly available at https://github.com/solids4foam/solids4foam on the `feature-petsc-snes` branch, and the cases and plotting scripts are available at https://github.com/solids4foam/solid-benchmarks.

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Appendix A WIP

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